

Exploring LSTM Proficiency in Multivariable Stock Price Prediction

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CIS4930 – Introduction to Machine Learning

Spring 2024

Abstract

In recent years, the use of Long Short-Term Memory (LSTM) neural networks has gained significant traction in the field of financial forecasting. Coupled with the increase in GPU processing power, many financial firms are now able to exploit quantitative techniques to analyze market conditions and execute automated trades. This leaves the task of predicting stock prices accurately to be a multibillion-dollar endeavor. This study aims to analyze the effectiveness of LSTM models in predicting stock prices by leveraging and engineering multivariable datasets, in hopes of gaining insight into their performance in handling diverse market indicators.

Background

Long Short-Term Memory models are a relatively new idea in the advancing field of Machine Learning, with their publishing paper, “Long Short-Term Memory” [1] being released in 1997. The goal of this type of neural network was to resolve the problem of exploding and vanishing error backflows caused by backpropagation in real-time recurrent learning [1]. The designers achieved this through the construction of novel memory cells to facilitate the constant error backpropagation. This allows the model to analyze and bridge very long time-lags in data patterns [1], a strong reason why the use of it in stock predictions, where the market is a predictable series of rises and falls throughout time has piqued the interest of so many researchers.

The basis of this project was built on “Forecasting the volatility of stock price index: A hybrid model integrating LSTM with multiple GARCH-type models” [2], where the LSTM model was used as part of a hybrid model combined with statistical models which aimed to forecast volatility and its impact on the stock prices. The findings of this article proved LSTM’s success in forecasting market

conditions when combined with various indicators as input, and showed ways which the predictions may be enhanced through various statistical model combinations. Specifically, this research analyzed the varying effect of several different types of generalized autoregressive conditional heteroscedasticity (GARCH) models in order to capture the volatility forecast, due to the fat-tail shaped probability distribution of financial time-series data. Volatility is an important market indicator which measures the current fluctuation in prices and therefore the degree of uncertainty or risk in an investment [2].

Methodology

Seeing the added benefit of using traditional, statistical market indicators in conjunction with a deep neural network, this project incorporated volatility prediction using a GARCH(2,1) model. Furthermore, moving average, which indicates the general direction that time-series data is trending in (positive/negative and by how much), was also incorporated with intervals of 10-Day, 20-Day, and 30-Day averages. The effect of the different interval averages plays a role in prediction window forecasting, as well as the level of accuracy we can forecast at, with 10-

Day averages being the most specific, and 30-Day averages being just the general direction, and therefore less fitted to the price curve. Additionally, the daily returns were calculated by:

$$Return = \frac{Close - Open}{Open}$$

where open represents the opening price for that day, close represents the closing price, therefore being a sort of calculation of percent change.

Experiment

With the goal of exploiting LSTM models' ability to bridge time-lags in data patterns, this project aimed to choose an industry represented by seasonal fluctuations which in turn affect the stock price. Through this scope, the entertainment industry was chosen, which has been shown to have an observed seasonality, especially in terms of movie revenue [3]. Specifically, stock tickers CMCSA, DIS, FOXA, and NFLX, corresponding to Comcast Corp. (owner of NBC and other production companies), Disney, Fox, and Netflix, respectively.

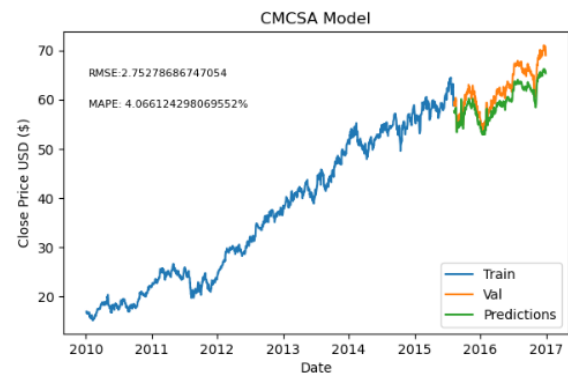
Upon successful engineering of the data, where volatility was predicted using the previously described GARCH(2,1) model, and the 10-, 20-, and 30-Day moving averages were computed, an LSTM model was trained by using Tensorflow's Sequential architecture. The LSTM Model had several layers, consisting of:

- LSTM layer with 256 memory cells.
- LSTM layer with 128 memory cells.
- LSTM layer with 64 memory cells.
- LSTM layer with 32 memory cells.
- Flatten, to reduce dimensionality of the data from previous layers for a smooth transition to the fully connected Dense layers.
- Dropout, for regularization and reduced risk of overfitting.
- Dense with 16 neurons.
- Dense with 1 neuron, outputs prediction.

The first test done with the model was to forecast prices for each ticker using input data from that day. This step was done to verify the validity of using an LSTM model at all for this type of regressive prediction, but also in order to observe the changes in fine tuning the model's hyperparameters and layers.

Next, this experiment aimed to analyze LSTM's effectiveness to predict prices at varying points in the future. This is the crux of a good prediction method, and at the core of the needs of financial institutions, so several windows of predictions were used, including 10-Day, 20-Day, 30-Day, 60-Day, and 120-Day. This opportunity also allowed for analysis on the effects of moving average window and prediction window on the accuracy of LSTM predictions.

Output for the experiments centered around line graphs representing the stock price for the training set, stock price for the test set, and the actual prediction overlayed, as shown in Figure 1.



1: Example output of LSTM prediction.

Additionally, metrics such as root mean-squared error (RMSE) and mean absolute percentage error (MAPE) were calculated for each prediction to analyze the models accuracy in prediction and the percentage in which the model was off, perhaps a more useful metric in the realm of stock predictions, where prices can range from hundreds of dollars to several dollars.

Results

The experiment garnered results which can be viewed from several different points of view.

Results by Ticker

The first lens which the results can be analyzed through is by the actual tickers themselves. Analyzing the output, we see that in the case of fully informed forecasting (model not trying to predict in the future, but rather through given data of that day), we see that FOXA and DIS outperform the other two tickers by a significant margin [Figure 2], with RMSE of 0.79 and 1.37, and MAPE at 2.76 and 1.16, respectively. This can be hypothesized by the persistent seasonality in the data, which fluctuations can be seen as consistent and symmetrical. This highlights not only LSTM models' ability to analyze time-series dependent trends, but also the role that the 30-Day moving average plays in the lag of predictions and the upward trend they produce. Furthermore, it is noted that more volatile data such as CMCSA and NFLX has a lower prediction ability, due to increased fluctuations, producing noise in the data and highly variable volatility numbers. It can also be noted that the volatility and 30-Day moving average estimate work against each other, with the 30-Day moving average being slow to react to changes, despite volatility measures highlighting them.

Results by Prediction Window

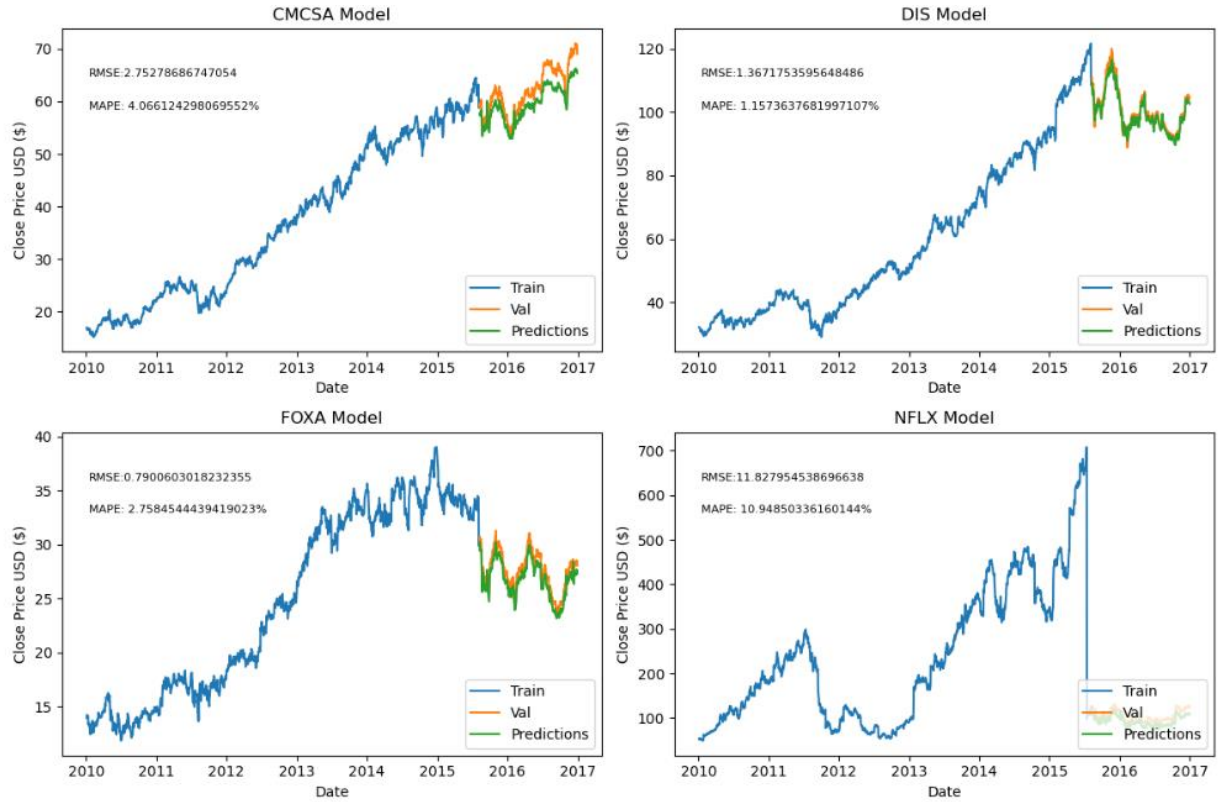
Further analysis can be done on the second half of the experiment through the lens of the predict-ahead window, which ranges from 10 to 120 days ahead. Results in this section were highly variable, however when speaking generally, it is noted that the 10-day prediction window yielded the best results. In the case of CMCSA, where data is more volatile, however patterns are produced in intervals of approximately 30 days, it is noted that the 30-Day interval was actually the most accurate, and the results of the 120-Day interval were second most accurate. It is generally seen in the data

that the accuracy increased with the reduction of prediction window size, as can be expected, but in rare cases, the 120-Day prediction window produced unexpectedly good results, a promising result of LSTM pattern recognition despite wide time gaps between the rise and fall data. The role that the moving average plays in the window prediction is not to go unnoted. In theory, the 30-Day moving average will be the least conforming to the line price, with the 10-Day moving average responding most to the noise in price fluctuation. One can clearly see this same pattern produced in the forecast windows, where the windows (10 and 20) that use the 10 and 20 day moving average fluctuate more closely with the data, and the windows using the 30-Day moving average, while still accounting for data fluctuations with precision, are much more generalized curves [Figure 3 vs Figure 4].

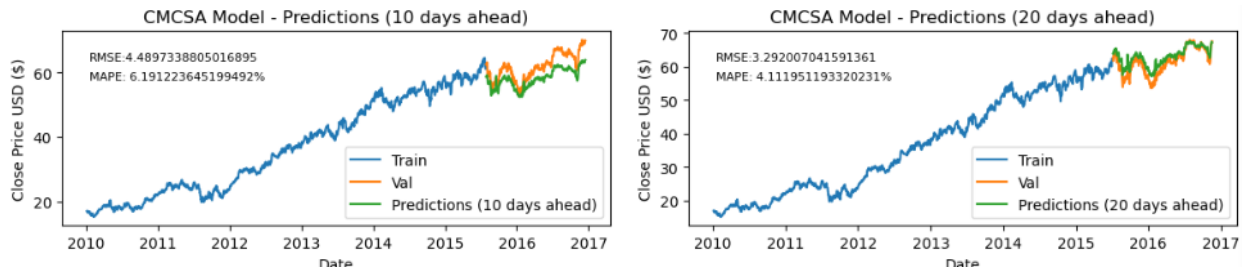
Conclusion

In summation, the result of this experiment proved promising for the use of LSTM in stock price prediction. The level of customizability in the model through hybrid models, like in [2], or through data engineering, like in this experiment, allow for the fine tuning of the model and a theoretical quantitative approach to be taken. The result went on to demonstrate the role which economic indicators such as volatility and moving average play in price forecasting, as well as the technicality behind these statistical flags and the impact which fine tuning these can have on the model.

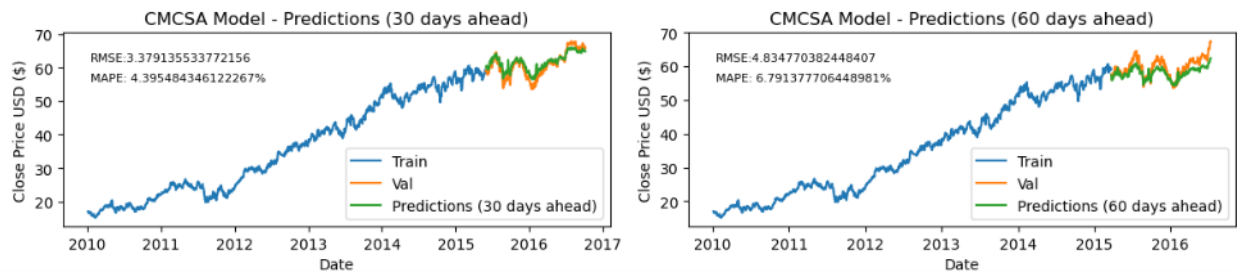
Appendix



2: Relationship between different tickers in their immediate forecast implementation.



3: Relationship between different prediction windows, in regards to line fitting.



4: Same as Figure 3, but for 30 and 60 day Windows

References

- [1] S. Hochreiter and J. Schmidhuber, "Long Short-Term Memory," 1997.
- [2] H. Y. Kim and C. H. Won, "Forecasting the volatility of stock price index: A hybrid model integrating LSTM with multiple GARCH-type models," 12 March 2018. [Online]. Available: <https://doi.org/10.1016/j.eswa.2018.03.002>. [Accessed 20 April 2024].
- [3] L. Einav, "Seasonality in the U.S. Motion Picture Industry," *The RAND Journal of Economics*, vol. 38, no. 1, pp. 127-145, 2007.